

Every-Iterate Quadratic Amplitudes for Finite Cyclic Rotations

Route-A structural certificate C161: baseline manuscript

Abstract

For an odd cyclic rotation with a quadratic observable, we reduce every complete Birkhoff amplitude to an odd quadratic Gauss sum. The baseline manuscript proves an exact greatest-common-divisor vanishing condition and an explicit nonzero evaluation. It also records the rejection of an earlier Heisenberg candidate whose local equivalences were not proved. Prime zero-level specialization and operator-symmetry receipts are deferred.

Keywords: cyclic dynamics; Birkhoff sum; quadratic Gauss sum; exact vanishing; model pivot.

中文摘要：本文的基线稿研究奇数阶循环旋转及二次观测测量，把任意迭代的完整振幅约化为有限二次高斯和，并证明最大公因数消失条件及非零分支求值。此前海森堡候选的局部等价尚未严格闭合，故不作为结果。零水平特例与算子对称性留待后续轮次。

关键词：循环动力学；比尔科夫和；二次高斯和；精确消失；模型切换。

1 Baseline all-iterate theorem

Let q be odd, $n \geq 1$, $R_q(x) = x + 1$, and $\phi(x) = ax^2 + bx$. Then

$$S_n \phi(x) = A_n x^2 + B_n x + C_n, \quad A_n = an, \quad B_n = an(n-1) + bn, \\ C_n = \frac{an(n-1)(2n-1)}{6} + \frac{bn(n-1)}{2}.$$

For $G_{q,n} = \sum_{x \bmod q} e^{2\pi i S_n \phi(x)/q}$, put $d = (A_n, q)$ and $Q = q/d$. Translation by Q proves

$$G_{q,n} = 0 \iff d \nmid B_n.$$

If $d \mid B_n$ and $Q > 1$, with $A' = A_n/d$, $B' = B_n/d$ and $h = -(4A')^{-1}(B')^2 \bmod Q$,

$$G_{q,n} = d \left(\frac{A'}{Q} \right) \epsilon_Q \sqrt{Q} e^{2\pi i (C_n + dh)/q}.$$

This follows by reduction to Q , completion of the square, and the primitive odd Gauss lemma. If $Q = 1$, the constant branch is $qe^{2\pi i C_n/q}$.

2 Gate and boundary

The initial Heisenberg continuation was rejected because quotient-coordinate, translation-removal, and 2- and 5-adic equivalences were not proved. This finite cyclic pivot makes an all-parameter source theorem. It does not produce a target trace law, Euler factor, or Hilbert–Pólya operator.

Declarations

All data and code are packaged; no external dataset is used. No human or animal subjects are involved. No competing interest or external funding is reported. An AI coding assistant supported drafting and code review; it was not treated as an external peer reviewer.